

NAME : B.Chaithanya Prakash Reddy

REG NO : 192425255

Q6. Debugging Bubble Sort Code

The following Bubble Sort implementation does not correctly sort the array and performs unnecessary iterations. Identify logical errors in loop conditions and swapping mechanism, and explain their causes. Debug the code step-by-step to ensure correct sorting. Optimize the algorithm by reducing unnecessary passes using a flag. Analyze the time complexity of the corrected version. Rewrite the final optimized code with proper comments.

```
def bubble_sort(arr):  
  
    n = len(arr)  
  
    for i in range(n):  
  
        for j in range(n-1):  
  
            if arr[j] &lt; arr[j+1]: # ERROR  
  
                arr[j], arr[j+1] = arr[j+1], arr[j]  
  
    return arr
```

```
main.py  [ ] [ ] [ ] Share Run Output Clear  
1 def bubble_sort(arr):  
2     n = len(arr)  
3     for i in range(n):  
4         for j in range(n-1):  
5             if arr[j] < arr[j+1]: # ERROR  
6                 arr[j], arr[j+1] = arr[j+1], arr[j]  
7     return arr  
  
ERROR!  
Traceback (most recent call last):  
  File "<main.py>", line 2  
    n = len(arr)  
    ^  
IndentationError: expected an indented block after function definition on line 1  
  
=== Code Exited With Errors ===
```

Error 1: Wrong Comparison Operator

```
if arr[j] < arr[j+1]:
```

Correction:

```
if arr[j] > arr[j+1]:
```

Problem:

- This swaps elements when the left element is **smaller** than the right.

- It sorts the array in **descending order** instead of ascending order.

main.py		Output	
<pre> 1 def bubble_sort(arr): 2 n = len(arr) 3 for i in range(n): 4 for j in range(n-1): 5 if arr[j] < arr[j+1]: # ERROR 6 arr[j], arr[j+1] = arr[j+1], arr[j] 7 return arr </pre>		<pre> ERROR! Traceback (most recent call last): File "<main.py>", line 2 n = len(arr) ^ IndentationError: expected an indented block after function definition on line 1 </pre> === Code Exited With Errors ===	

Error 2: Unnecessary Inner Loop Iterations

for j in range(n-1):

Problem:

- After every pass, the largest element reaches its correct position.
- Comparing it again is unnecessary.

Correction:

for j in range(n-i-1):

This reduces unnecessary comparisons.

main.py		Output	
<pre> 1 def bubble_sort(arr): 2 n = len(arr) 3 for i in range(n): 4 for j in range(n-i-1): 5 if arr[j] > arr[j+1]: # ERROR 6 arr[j], arr[j+1] = arr[j+1], arr[j] 7 return arr </pre>		<pre> ERROR! Traceback (most recent call last): File "<main.py>", line 2 n = len(arr) ^ IndentationError: expected an indented block after function definition on line 1 </pre> === Code Exited With Errors ===	

Error 3: No Early Stopping

The algorithm continues even if the array is already sorted.

Solution :

Use a **flag**.

If no swaps occur in one pass:

swapped = False

After swapping:

swapped = True

If

if not swapped:

break

the algorithm stops early.

2. Step-by-Step Debugging

Example:

arr = [5, 2, 4, 1]

Pass 1

Compare 5 and 2

Swap

[2,5,4,1]

Compare 5 and 4

Swap

[2,4,5,1]

Compare 5 and 1

Swap

[2,4,1,5]

Largest element **5** is fixed.

Pass 2

Compare 2 and 4

No swap

[2,4,1,5]

Compare 4 and 1

Swap

[2,1,4,5]

Largest unsorted element **4** is fixed.

Pass 3

Compare 2 and 1

Swap

[1,2,4,5]

Array becomes sorted.

Pass 4

No swaps occur.

Algorithm stops because of the flag.

3. Optimized Bubble Sort Code

```
def bubble_sort(arr):  
    n = len(arr)  
    # Traverse through all array elements  
    for i in range(n):  
        # Flag to check if any swapping happened  
        swapped = False  
        # Last i elements are already in correct position  
        for j in range(n - i - 1):  
            # Swap if left element is greater than right element  
            if arr[j] > arr[j + 1]:  
                arr[j], arr[j + 1] = arr[j + 1], arr[j]  
                swapped = True  
        # If no swapping occurred, array is already sorted  
        if not swapped:
```

```

        break

    return arr

# Example

arr = [5, 2, 4, 1]

print("Sorted Array:", bubble_sort(arr))

```

Output :

Sorted Array: [1, 2, 4, 5]

The screenshot shows a code editor with a file named 'main.py'. The code defines a 'bubble_sort' function that sorts an array in ascending order. It includes comments for each step: traversing elements, checking for swaps, and breaking the loop if the array is already sorted. An example array [5, 2, 4, 1] is used to demonstrate the function. The output panel on the right shows the sorted array [1, 2, 4, 5] and a success message.

```

main.py
1- def bubble_sort(arr):
2     n = len(arr)
3
4     # Traverse through all array elements
5-     for i in range(n):
6
7         # Flag to check if any swapping happened
8         swapped = False
9
10        # Last i elements are already in correct position
11-        for j in range(n - i - 1):
12
13            # Swap if left element is greater than right element
14-            if arr[j] > arr[j + 1]:
15                arr[j], arr[j + 1] = arr[j + 1], arr[j]
16                swapped = True
17
18        # If no swapping occurred, array is already sorted
19-        if not swapped:
20            break
21
22    return arr
23
24
25 # Example
26 arr = [5, 2, 4, 1]
27 print("Sorted Array:", bubble_sort(arr))

```

Output

Sorted Array: [1, 2, 4, 5]

=== Code Execution Successful ===

5. Time Complexity Analysis

Case	Time Complexity
Best Case (already sorted)	$O(n)$
Average Case	$O(n^2)$

Case	Time Complexity
Worst Case (reverse sorted)	$O(n^2)$

Space Complexity

$O(1)$ (In-place sorting)

Optimization Summary

- Corrected comparison from $<$ to $>$ for ascending order.
- Reduced unnecessary comparisons using $\text{range}(n - i - 1)$.
- Added a swapped flag to stop early when the array is already sorted.
- Improved best-case performance from $O(n^2)$ to $O(n)$ while keeping the average and worst cases at $O(n^2)$.

Algorithm: Optimized Bubble Sort

1. **Start.**
2. Read the input array `arr` of size n .
3. Repeat for each pass from $i = 0$ to $n-1$:
 - Initialize a Boolean variable `swapped = False`.
4. Compare each pair of adjacent elements from index 0 to $n-i-2$.
5. If `arr[j] > arr[j+1]`, swap the two elements.
6. After swapping, set `swapped = True`.
7. Continue until the end of the current pass.
8. If `swapped == False`, terminate the algorithm because the array is already sorted.

9. Otherwise, continue with the next pass.
10. Repeat until all passes are completed or the array becomes sorted.
11. Display the sorted array.
12. **Stop.**